BAD702 – Statistical & Machine Learning

Exam-ready answers (10 marks each) • Based only on Module 3, 4 and 5 textbook notes

Q1. Explain random assignment and permutation testing in A/B experiments. State H_0. In a two-sided test, 19 of 999 shuffles are at least as extreme as observed. Find p = (19 + 1)/(999 + 1). Decide at α = 0.05 before and after Bonferroni correction for four tests. [10 Marks]

1. A/B test and random assignment

An **A/B test** is an experiment with two groups to establish which of two treatments is superior. The **control group** gets the standard (or no) treatment and the **treatment group** gets the new one. The **test statistic** (e.g. click rate, mean time) is chosen *before* the experiment to avoid researcher bias.

Randomization means subjects are assigned to treatments randomly. Then any difference between the groups is due to only one of two things: (i) the real effect of the treatment, or (ii) luck of the draw in assignment (chance).

2. Hypotheses

- **H_0 (null hypothesis):** the treatments are equivalent; there is no difference between A and B and any observed difference is due to chance.
- **H_1 (alternative hypothesis):** A is different from B (bidirectional). A **two-way (two-tail) test** counts extreme chance results in both directions.

3. Permutation test procedure

1. Combine the results of groups A and B into a single data set (this embodies H_0: groups are interchangeable).
2. Shuffle and randomly draw (without replacement) a resample of size n_A, then one of size n_B from the remaining data.
3. Calculate the test statistic (e.g. difference in means/proportions) for the resamples – one permutation iteration.
4. Repeat R times to get the **permutation distribution**.
5. Compare the observed difference with this distribution. If it lies well within it, the difference is chance; if it lies outside most of it, the result is **statistically significant**.

4. p-value calculation

The **p-value** is the probability that a chance model (H_0) produces a result as unusual or extreme as the observed result.

$$p = (19 + 1) / (999 + 1) = 20 / 1000 = 0.02$$

5. Decision

Case	α	Comparison	Decision
Before Bonferroni	0.05	0.02 < 0.05	Reject H_0 – significant
After Bonferroni (4 tests)	0.05 / 4 = 0.0125	0.02 > 0.0125	Fail to reject H_0 – not significant

Bonferroni adjustment: running many tests on the same data increases the chance of a Type 1 error (**alpha inflation**). Bonferroni divides α by the number of comparisons, giving a stricter bar for each test.

Conclusion

The result is significant for a single test, but not significant after correcting for four tests.

Q2. For 16 users, paired time reductions average 3 s with SD 4 s. Assume independent normal differences. State hypotheses for a positive mean reduction. Find t and df. Decide at α = 0.05 (critical t = 1.753) and justify the paired test. [10 Marks]

Given

n = 16, mean difference $\bar{d}$ = 3 s, SD = 4 s, α = 0.05, critical t = 1.753.

1. Hypotheses (one-tail, positive mean reduction)

- **H_0:** mean reduction ≤ 0 (no improvement; difference is due to chance).
- **H_1:** mean reduction > 0 (time is really reduced).

2. t-statistic

The **t-statistic** is a standardized test statistic compared with the t-distribution (the reference distribution under H_0).

$$t = \bar{d} / (SD / \sqrt{n}) = 3 / (4 / \sqrt{16}) = 3 / 1 = 3$$

3. Degrees of freedom

$$df = n - 1 = 16 - 1 = 15$$

Degrees of freedom is the number of values free to vary; it selects the correct t-distribution.

4. Decision

Calculated t = 3 > critical t = 1.753, so **reject H_0**. The mean reduction is statistically significant at α = 0.05.

5. Justification of the paired test

- The same 16 users are measured twice (before and after), so the two sets of values are not independent.
- Analysing each user's **difference** removes user-to-user variation, so only the effect of the change remains.
- This makes the test more sensitive. The t-test assumes the differences are independent and normal (given in the question).

Conclusion

The new design gives a significant reduction in time. (Textbook note: the p-value is only one piece of information bearing on a decision.)

Q3. Explain reference coding using $\hat{y} = 20 + 3x + 8D$ (store B: D = 1; store A: D = 0). Interpret 3 and 8. Predict for B at x = 2. Suggest a polynomial remedy for U-shaped residuals. In A/B testing, how does a smaller effect change sample size at fixed α, power and variability? [10 Marks]

1. Reference coding

Regression needs numeric inputs, so a **factor (categorical) variable** is recoded into **dummy variables** (binary 0/1). In **reference coding** (treatment coding) one level is the **reference** and every other level is compared with it. A factor with P levels needs **P − 1 dummies**. Here store A is the reference (D = 0) and store B is coded D = 1.

2. Interpretation of $\hat{y} = 20 + 3x + 8D$

- **3 (slope of x):** for every one-unit increase in x, ŷ increases by 3, holding the store constant.
- **8 (coefficient of D):** store B has a predicted value 8 units higher than reference store A at the same x.
- **20 (intercept):** predicted value for store A when x = 0.

3. Prediction for store B at x = 2

$$\hat{y} = 20 + 3(2) + 8(1) = 20 + 6 + 8 = 34$$

4. Polynomial remedy for U-shaped residuals

U-shaped residuals (seen in a **partial residual plot**) show the relationship is nonlinear. Use **polynomial regression** by adding a squared term:

$$Y = b_0 + b_1X + b_2X^2 + e$$

The squared term captures the curvature. If still not flexible enough, use **splines** or a **GAM**.

5. Effect of a smaller effect size on sample size

The four moving parts are **sample size, effect size, significance level (α) and power**; fix any three and the fourth is calculated.

- With α, power and variability fixed, a **smaller effect needs a larger sample size**, because small differences are harder to separate from chance.
- Example: distinguishing a .300 hitter from a .280 hitter needs many more at-bats than .350 vs .200. In the ad example, a 10% boost needed far more impressions than a 50% boost.

Q4. An ε-greedy bandit compares three designs (ε = 0.10). Explain exploration and exploitation. Find expected exploratory visits out of 200. For $\hat{y} = 10 + 2x + 3\max(0, x - 5)$, predict at x = 7. Identify the spline's knot and slopes on either side. [10 Marks]

1. Multi-arm bandit

A **multi-arm bandit** is an imaginary slot machine with many arms, each paying at a different rate. An **arm** is a treatment (here a design) and a **win** is a click. Unlike a traditional A/B test, it shifts traffic to the better treatment *during* the experiment, reduces exposure to the inferior treatment, and handles 3 or more treatments efficiently.

2. ε-greedy algorithm

Generate a uniform random number between 0 and 1:

- **Exploration:** if the number is below ε, show a randomly chosen design. This keeps testing every design so a winner hidden by chance can emerge.
- **Exploitation:** if the number is ≥ ε, show the design with the highest response rate so far. This takes advantage of the apparent winner.

ε is the single governing parameter: ε = 1 gives a plain A/B test; ε = 0 gives a purely greedy algorithm.

3. Expected exploratory visits

$$\textbf{Exploratory visits} = \varepsilon \times \textbf{total visits} = 0.10 \times 200 = 20$$

The remaining 180 visits are exploitation.

4. Spline prediction at x = 7

$$\hat{y} = 10 + 2(7) + 3 \cdot \max(0, 7 - 5) = 10 + 14 + 6 = 30$$

5. Knot and slopes

A **knot** is the value that separates spline segments.

- **Knot:** x = 5 (where max(0, x − 5) starts to act).
- **Slope before the knot (x ≤ 5):** 2.
- **Slope after the knot (x > 5):** 2 + 3 = 5.

Q5. Explain within-class covariance and class separation in Fisher's linear discriminant. Contrast this with a GLM and name its three components. Select and justify response families and links for binary purchase outcomes and daily event counts. [10 Marks]

1. Fisher's linear discriminant

Fisher's discriminant separates the variation **between** groups from the variation **within** groups. For two groups (y = 0 and y = 1), LDA finds the linear combination $w_x x + w_z z$ that maximizes:

$$\mathbf{SS_{between} / SS_{within}}$$

- **Class separation (between sum of squares):** the squared distance between the two group means – should be large.
- **Within-class covariance (within sum of squares):** the spread of records around their own group mean, weighted by the **covariance matrix** (variances on the diagonal, covariances off the diagonal) – should be small.

Maximizing the ratio gives the greatest separation. The result is the **discriminant function**; its scores (**discriminant weights**) estimate class probabilities. LDA assumes normally distributed predictors and the same covariance matrix for both classes.

2. Contrast with GLM (e.g. logistic regression)

LDA	GLM
Uses covariance matrix and between/within sums of squares	Fitted by **maximum likelihood estimation** (deviance)
Assumes normally distributed predictors	No normality assumption for predictors
Gives discriminant scores	Models log odds, mapped to probability by the link

3. Components of a GLM

1. **Probability distribution (family)** of the response.
2. **Link function** – transformation mapping the response to the predictors.
3. **Linear predictor** ($\beta_0 + \beta_1 x_1 + \ldots + \beta_q x_q$) on which the link acts. (The textbook states the first two explicitly; the linear predictor is the third part.)

4. Choice of family and link

Outcome	Family	Link	Justification
Binary purchase (buy / don't buy)	**Binomial**	**Logit** (log odds)	Outcome is 0/1; logit keeps probability between 0 and 1
Daily event counts	**Poisson**	**Log**	Poisson is commonly used for count data (e.g. visits in a period)

Q6. A logistic regression model performs poorly. You suspect nonlinear effects for a predictor X. Demonstrate how spline regression could be added to improve the model. [10 Marks]

1. Problem

Logistic regression assumes the **log odds** is linear in each predictor. If predictor X has a nonlinear effect, the model fits poorly.

2. Steps to add spline regression

1. **Detect nonlinearity:** plot the **partial residuals** for X; they are useful to confirm nonlinear behaviour and identify influential records. If the linear fit does not follow the smooth curve, X is nonlinear.
2. **Why not only polynomials:** they capture limited curvature, and higher-order terms cause undesirable "wiggliness".
3. **Use a spline:** a series of piecewise continuous polynomials joined smoothly at fixed points called **knots**. Specify the **degree** (e.g. cubic = 3) and **knot locations** (e.g. quartiles).
4. **Fit within logistic regression** (GLM with Binomial family):
5. **Evaluate** using classification metrics (confusion matrix, AUC, lift) on holdout data.

Python (statsmodels):

```
formula = 'outcome ~ bs(payment_inc_ratio, df=4) + purpose_ + home_ + emp_len_ +
bs(borrower_score, df=4)'
model = smf.glm(formula=formula, data=loan_data, family=sm.families.Binomial())
```

R (GAM, mgcv): the term s(X) selects the best knots automatically:

```
gam(outcome ~ s(payment_inc_ratio) + purpose_ + home_ + emp_len_ + s(borrower_score),
data=loan_data, family='binomial')
```

3. Points to note

- Spline coefficients are **not directly interpretable**; use the visual (partial residual) display.
- A better fit is not automatically a better model – check that the result makes sense (confounding variables).

Conclusion

Splines let the log odds bend with X, capturing the nonlinear effect and improving the logistic model.

Q12. A logistic regression model performs poorly. You suspect nonlinear effects for a predictor X. Describe how spline regression could be added to improve the model. [10 Marks]

Answer: This is the same question as Q6 (only the verb differs). Write the answer given under Q6 – the problem, steps 1 to 5, the two code lines, and the conclusion.

Q7. Given an observed difference of 0.015 and 10,000 permutation samples, calculate the p-value if 2,700 samples produced differences greater than the observed value. [10 Marks]

Method

The p-value is the proportion of times the chance model (H_0) produces a difference equal to or more extreme than the observed one.

$$p = 2,700 / 10,000 = 0.27$$

Interpretation

- p = 0.27 means a result this extreme would occur about **27% of the time by random chance**.
- At α = 0.05, 0.27 > 0.05, so **fail to reject H_0**.
- The observed difference 0.015 is well within the range of chance variation – **not statistically significant**.

Textbook note: a non-significant p-value means “effect not proven”, not that no effect exists; a larger sample might give a smaller p-value.

Q8. Given a sample size of 1,00,000 patients where a new treatment improves success by 0.1%, determine whether this difference is likely statistically significant and discuss its practical significance. [10 Marks]

1. Statistical significance

A result is **statistically significant** if it is more extreme than what chance variation can produce (small p-value).

- With a very large sample (1,00,000), sampling variability is small, so even a tiny difference can be big enough to **rule out chance**.
- So the 0.1% improvement is **likely to be statistically significant**.
- Caution (textbook): the sample size needed depends on the number of actual successes, not only total records, so confirm with a test and its p-value.

2. Practical significance

- A 0.1% gain is about **1 extra success per 1,000 patients** (about 100 in 1,00,000) – a very small effect.
- Even if significant, it may be unimportant: ruling out chance does not make an unimportant result important.
- Practical value depends on cost, side effects and benefit of the treatment.

3. Related points

- A p-value does not measure the **size of an effect** or importance of a result (ASA principle 5).
- Decisions should not rest only on whether a p-value passes a threshold.

Conclusion

The difference is probably statistically significant because of the huge sample, but it may have little practical significance unless 100 extra successes justify the treatment's cost and risk.

Q9. Given a categorical variable with 4 levels, apply the rules to determine how many dummy variables are needed. Apply interpretation rules to explain what a negative coefficient for one category (as compared to the reference) means. [10 Marks]

1. Rule

A factor (categorical) variable with P distinct levels is represented by **P − 1 dummy (binary 0/1) variables**, using **reference coding**.

$$\mathbf{P = 4 \rightarrow 4 - 1 = 3 \text{ dummy variables}}$$

2. Why not 4

Including all 4 levels (one hot encoding) makes the dummies perfectly redundant: if the first 3 are 0, the 4th must be 1. This causes a **multicollinearity** error in regression. Only 3 are free to vary (degrees of freedom idea).

3. Example

Levels A, B, C, D with D as reference: create dummies for A, B and C. The omitted level D is the **reference level**.

4. Meaning of a negative coefficient

- In reference coding every coefficient compares its level with the **reference level**.
- A **negative coefficient** for category B means that, **compared with the reference category, B has a lower predicted value of the outcome**, holding other variables constant.
- The size is the amount lower, e.g. −5 means 5 units lower than the reference.

Q10. Given a polynomial model of degree 3, explain what happens to the model shape when one squared term is removed. [10 Marks]

1. Cubic polynomial model

Polynomial regression adds powers of a predictor to the regression equation:

$$\mathbf{Y = b_0 + b_1X + b_2X^2 + b_3X^3 + e}$$

- X term – straight-line trend.
- X^2 term – basic curvature (bend / U-shape).
- X^3 term – an additional change of curvature (S-shape).

2. After removing the squared term

$$\mathbf{Y = b_0 + b_1X + b_3X^3 + e}$$

- The **quadratic bend is lost**; the model can no longer fit simple U-shaped or hill-shaped curvature.
- The curve is controlled only by the linear and cubic terms, so its shape is more restricted and may fit the data less well.
- The fitted curve changes, so residuals should be rechecked with a **partial residual plot**.

3. Remember

Higher-order terms can cause “wiggliness”; for flexible fits, **splines** or a **GAM** are better than adding or removing terms blindly.

Note: the textbook does not discuss this exact case; the answer follows from the polynomial form given there.

Q11. A logistic regression coefficient for "age" is 0.09. Apply the odds ratio interpretation to explain its effect on the likelihood of disease. [10 Marks]

1. Key idea

In logistic regression the coefficient β_i is the **log of the odds ratio** for X_i.

$$\mathbf{\log(Odds(Y = 1)) = \beta_0 + \beta_1x_1 + \ldots + \beta_qx_q}$$

Odds = p / (1 − p), the ratio of “success” (disease) to “not success”.

2. Calculation

$$\mathbf{\text{Odds ratio} = e^{\beta} = e^{0.09} \approx 1.09}$$

3. Interpretation

- For each **one-unit (one-year) increase in age**, the odds of disease are multiplied by about 1.09, i.e. increase by about **9%**, holding other variables constant.
- The coefficient is **positive**, so the odds ratio is greater than 1 – **older age increases the likelihood of disease**.
- The effect compounds: for a 10-year increase, odds ratio = $e^{0.9} \approx 2.46$.
- Odds ratio = 1 means no effect; below 1 means lower odds.

Textbook parallel: a coefficient of 0.08244 for payment-to-income ratio gives exp(0.08244) ≈ 1.09 times higher odds of default.

Conclusion

Age is a risk factor: each extra year raises the odds of disease by roughly 9%.